

Zijian (Longino) ZHAO 赵子健

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Education

The Hong Kong University of Science and Technology (Clearwater Bay Campus, Hong Kong)
Ph.D. in Civil Engineering (Scientific Computation)
Supervisor: Prof. Sen Li
Research Directions: Reinforcement Learning, LLMs, Operations Research, Intelligent Transportation, AI Ethics

Sep. 2024 – Present
GPA: 3.8/4.2, Rank: N/A

Sun Yat-Sen University (Guangzhou Campus)
B.Eng. in Computer Science and Technology (National Basic Subject Talent Training Plan)
Supervisors: Prof. Kai Huang, Prof. Chengying Gao, Prof. Ning Liu

Sep. 2020 – Jul. 2024
GPA: 4.0/5.0, Rank: Top 10%

Research Directions: Domain Adaption, Wireless Sensing, Music Information Retrieval, Robot Control
Thesis: Motion Control Method for Small Quadruped Robots Based on Trajectory Modulation and Reinforcement Learning
Rank First in: Computer Programming, Principles of Compilers, Distributed Systems, Embedded Systems, Complex Variables, Mathematical Analysis, Advanced Algebra, Data Structures and Algorithms, Probability and Statistics, Discrete Mathematics

Experience

Research Intern
Noah's Ark Lab, 2012 Laboratories, Huawei

Apr. 2026 – Oct. 2026
Mentor: Xialiang Tong

Industry-Academia-Research Student
Likelihood Lab, Guangzhou WenSight Intelligent Technology Co., Ltd.

Feb. 2024 – Aug. 2024
Online, Part-time

Visiting Student
Shenzhen Research Institute of Big Data, The Chinese University of Hong Kong (Shenzhen)

Aug. 2023 – Aug. 2024
Host: Prof. Guangxu Zhu

Research Interest

My research interests lie in sensing, prediction, and decision management for human-centered smart cities and intelligent transportation systems, addressing problems such as order dispatch, algorithmic labor management, wireless sensing, and traffic prediction. On the technical side, I focus primarily on deep learning, multi-agent reinforcement learning, transfer learning, federated learning, and large models. Overall, I am interested in and open to interdisciplinary AI, applied AI, and AI humanities.

Publications

*: Corresponding Author; #: Project Lead; ¹: Equal Contributions; [J/C/R/P]: Journal/Conference/Report/Working Preprints

A. Selected Papers:

- [C1] **Zijian Zhao**, Sen Li*, “Triple-BERT: Do We Really Need MARL for Ride-Sharing Order Dispatch?”, 2026 International Conference on Learning Representations (ICLR), 2026 (CCF-A, **oral**) [[Paper](#)] [[Code](#)]
- [C2] **Zijian Zhao**, Dian Jin, Zijing Zhou, Xiaoyu Zhang*, “Automatic Stage Lighting Control: Is it a Rule-Driven Process or a Generative Task?”, 2026 International Conference on Learning Representations (ICLR), 2026 (CCF-A) [[Paper](#)] [[Code](#)] [[Dataset](#)] [[Model](#)]
- [J1] **Zijian Zhao**, Sen Li*, “The Impacts of Data Privacy Regulations on Food-Delivery Platforms”, Transportation Research Part C: Emerging Technologies (TR_C), 2025 (JCR-Q1) [[Paper](#)] [[Code](#)]
- [J2] **Zijian Zhao**, Sen Li*, “Discriminatory Order Assignment and Payment-Setting of On-Demand Food-Delivery Platforms: A Multi-Action and Multi-Agent Reinforcement Learning Framework”, Transportation Research Part E: Logistics and Transportation Review (TR_E), 2025 (JCR-Q1, ABS-3) [[Paper](#)] [[Code](#)]
- [J3] **Zijian Zhao**, Fanyi Meng, Zhonghao Lyu, Hang Li, Xiaoyang Li, Guangxu Zhu*, “CSI-BERT2: A BERT-Inspired Framework for Efficient CSI Prediction and Recognition in Wireless Communication and Sensing”, IEEE Transactions on Mobile Computing (TMC), 2025 (JCR-Q1, CCF-A) [[Paper](#)] [[Code](#)] [[Dataset](#)] [[Model](#)]
- [J4] **Zijian Zhao**, Zhijie Cai, Tingwei Chen, Xiaoyang Li, Hang Li, Qimei Chen, Guangxu Zhu*, “KNN-MMD: Cross Domain Wireless Sensing via Local Distribution Alignment”, IEEE Transactions on Mobile Computing (TMC), 2025 (JCR-Q1, CCF-A) [[Paper](#)] [[Code](#)] [[Dataset](#)]
- [J5] **Zijian Zhao**, Tingwei Chen, Zhijie Cai, Xiaoyang Li, Hang Li, Qimei Chen, Guangxu Zhu*, “CrossFi: A Cross Domain Wi-Fi Sensing Framework Based on Siamese Network”, IEEE Internet of Things Journal (IOT-J), 2025 (JCR-Q1, CCF-C) [[Paper](#)] [[Code](#)]

B. Other Papers:

- [C3] **Zijian Zhao**, “Towards Fairness in Transportation Gig Markets: Identifying, Imitating, and Mitigating Algorithm Discrimination via Deep Reinforcement Learning”, 2026 Annual AAAI Conference on Artificial Intelligence (AAAI) / Special Interest Group on Artificial Intelligence (SIGAI) Doctoral Consortium, 2026 (CCF-A DC) [[Paper](#)] [[Code](#)] [[Interview](#)]
- [C4] **Zijian Zhao***, Dian Jin, Zijing Zhou, “Zero-Effort Image-to-Music Generation: An Interpretable RAG-Based VLM Approach”, 2026 ACM International Conference on Multimedia Retrieval (ICMR), 2026 (CCF-B short paper) [[Paper](#)] [[Code](#)]
- [C5] **Zijian Zhao**, Sen Li*, “Discriminatory Order Assignment and Payment Setting on Food-Delivery Platforms: The Implicit Algorithmic Biases”, 2026 INFORMS Transportation Science and Logistics Conference (TSL), 2026 [[Paper](#)] [[Code](#)]
- [C6] **Zijian Zhao**, Jing Gao*, Sen Li, “Ride-Hailing Order Dispatching with A Mixture of On-Demand and Pre-Booked Requests via Reinforcement Learning”, 2026 COTA International Conference of Transportation Professionals (CICTP), 2026 [[Paper](#)] [[Code](#)]
- [C7] Shiting Chen¹, **Zijian Zhao**¹, Jinsong Chen*, “Confident RAG: Enhancing the Performance of LLMs for Mathematics Question Answering through Multi-Embedding and Confidence Scoring”, 2026 International Conference on Learning Representations (ICLR) LLM Reasoning Workshop, 2026 (CCF-A workshop) & 2026 International Conference on Computers in Education (ICCE), 2026 [[Paper](#)] [[Code](#)]
- [C8] Chuxue Cao, Jinluan Yang, Haoran Li, Kunhao Pan, **Zijian Zhao**, Zhengyu Chen, Yuchen Tian, Lijun Wu, Conghui He, Sirui Han, Yike Guo, “Pushing the Boundaries of Natural Reasoning: Interleaved Bonus from Formal-Logic Verification”, 2026 International Conference on Machine Learning (ICML), 2026 (CCF-A) [[Paper](#)] [[Model](#)]
- [C9] **Zijian Zhao***, “Let Network Decide What to Learn: Symbolic Music Understanding Model Based on Large-Scale Adversarial Pre-Training”, 2025 ACM International Conference on Multimedia Retrieval (ICMR), 2025 (CCF-B short paper) [[Paper](#)] [[Code](#)] [[Model](#)]
- [C10] **Zijian Zhao**, Tingwei Chen, Fanyi Meng, Zhijie Cai, Hang Li, Xiaoyang Li, Guangxu Zhu*, “LoFi: Vision-Aided Label Generator for Wi-Fi Localization and Tracking Sensing”, 2025 IEEE Global Communications Conference (GLOBECOM) GenAI NGN Workshop, 2025 (CCF-C workshop) [[Paper](#)] [[Code](#)] [[Dataset](#)]
- [C11] **Zijian Zhao**, Zhijie Cai, Tingwei Chen, Xiaoyang Li, Hang Li, Qimei Chen, Guangxu Zhu*, “Does MMD Really Align? A Cross Domain Wireless Sensing Method via Local Distribution”, 2025 IEEE/CIC International Conference on Communications in China (ICCC), 2025 [[Paper](#)] [[Code](#)]
- [C12] **Zijian Zhao**, Sen Li*, “Multi-Agent Reinforcement Learning for Order Assignment and Payment Setting on Food-Delivery Platforms: The Implicit Algorithmic Biases”, 2025 International Symposium on Transportation Data & Modeling (ISTDM), 2025 [[Paper](#)] [[Code](#)]
- [J6] Haolong Chen, Hanzhi Chen, **Zijian Zhao**, Kaifeng Han*, Guangxu Zhu*, Yichen Zhao, Ying Du, Wei Xu, Qingjiang Shi, “An Overview of Domain-Specific Foundation Model: Key Technologies, Applications and Challenges”, Science China Information Sciences (SCIS), 2025 (JCR-Q1, CCF-A) [[Paper](#)]
- [C13] Tingwei Chen, Yantao Wang, Hanzhi Chen, **Zijian Zhao**, Xinhao Li, Nicola Piovesan, Guangxu Zhu*, Qingjiang Shi, “Modeling the 5G Energy Consumption using Real-World Data: Energy Fingerprint is All You Need”, 2025 IEEE Global Communications Conference (GLOBECOM) GenAI NGN Workshop, 2025 (CCF-C workshop) [[Paper](#)] [[Code](#)]
- [C14] Chuxue Cao, Mengze Li, Juntao Dai, Jinluan Yang, **Zijian Zhao**, Shengyu Zhang, Weijie Shi, Chengzhong Liu, Sirui Han, Yike Guo, “Towards Advanced Mathematical Reasoning for LLMs via First-Order Logic Theorem Proving”, 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025 (CCF-B) [[Paper](#)] [[Code](#)]
- [C15] **Zijian Zhao**, Tingwei Chen, Fanyi Meng, Hang Li, Xiaoyang Li, Guangxu Zhu*, “Finding the Missing Data: A BERT-Inspired Approach Against Package Loss in Wireless Sensing”, 2024 IEEE International Conference on Computer Communications (INFOCOM) DeepWireless Workshop, 2024 (CCF-A workshop) [[Paper](#)] [[Code](#)] [[Dataset](#)] [[Model](#)]
- [C16] Xiao Liang, **Zijian Zhao**[#], Weichao Zeng, Yutong He, Fupeng He, Yiyi Wang, Chengying Gao*, “PianoBART: Symbolic Piano Music Understanding and Generating with Large-Scale Pre-Training”, 2024 IEEE Conference on Multimedia Expo (ICME), 2024 (CCF-B, **oral**) [[Paper](#)] [[Code](#)] [[Model](#)]
- [C17] Zitao Zhang, Yuhong Huang, **Zijian Zhao**, Zhenshan Bing, Chenglin Cai, Alois Knoll and Kai Huang*, “Autonomous Locomotion of a Rat Robot Based on Model-Free Reinforcement Learning”, 2024 IEEE International Conference on Advanced Robotics and Mechatronics (ICARM), 2024 [[Paper](#)]
- [C18] Zitao Zhang, Yuhong Huang, **Zijian Zhao**, Zhenshan Bing, Alois Knoll and Kai Huang*, “A Hierarchical Reinforcement Learning Approach for Adaptive Quadruped Locomotion of a Rat Robot”, 2023 IEEE International Conference on Robotics and Biomimetics (ROBIO), 2023 (**best paper finalist**) [[Paper](#)]
- [C19] Zitao Zhang*, Yuhong Huang, **Zijian Zhao**, Zhenshan Bing, Kai Huang, “Autonomous Locomotion of a Rat Robot Based on Reinforcement Learning”, 2023 CCF China Intelligent Robotics Annual Conference (CIRAC), 2023 [[Paper](#)]

C. Working Papers:

- [P1] **Zijian Zhao**, Sen Li*, “Beyond Multi-Agent Reinforcement Learning: Scalable Centralized Control for Large-Scale Dynamic Trip-Vehicle Assignment” (major revision, INFORMS Journal on Computing (IJOC))
- [P2] **Zijian Zhao**, Sen Li*, “OSPO: One-Step Policy Optimization for Real-Time Order Assignment on Ride-Sharing Platforms with Autonomous Vehicles” (major revision, Transportation Research Part C: Emerging Technologies (TR_C))
- [P3] **Zijian Zhao**, Jing Gao*, Sen Li, “Real-Time Order Assignment for Ride-Sharing Platforms with a Mixture of Pre-booked and On-Demand Requests” (minor revision, Transportation Research Part C: Emerging Technologies (TR_C))
- [P4] **Zijian Zhao**, Sen Li*, “Discriminatory Labor Management for On-Demand Meal Delivery Platforms with a Mixed Fleet of AVs and Human Couriers” (under review, Transportation Research Part C: Emerging Technologies (TR_C))

- [P5] **Zijian Zhao**, Yitong Shang, Sen Li*, “Personalized Federated Traffic Prediction via Prompt-Guided Global Representation and Node-Specific Adaptation”(under review, Transportation Research Part C: Emerging Technologies (TR_C))
- [P6] Yang Liu¹, **Zijian Zhao**¹, Yitong Shang, Sen Li*, “Joint Infrastructure Planning and Order Assignment for On-Demand Food-Delivery Services with Coordinated Drones and Human Couriers” (under review, Transportation Research Part C: Emerging Technologies (TR_C))
- [P7] **Zijian Zhao**, Sen Li*, “One Step is Enough: Multi-Agent Reinforcement Learning Based on One-Step Policy Optimization for Order Dispatch on Ride-Sharing Platforms” (under review, 2026 Annual Conference on Neural Information Processing Systems (NeurIPS)) [[ArXiv](#)] [[Code](#)]
- [P8] **Zijian Zhao**, Dian Jin, Zijing Zhou, Xiaoyu Zhang*, “Stage Light is Sequence²: Multi-Light Control via Imitation Learning” (under review, 2026 Annual Conference on Neural Information Processing Systems (NeurIPS)) [[ArXiv](#)] [[Code](#)]
- [P9] **Zijian Zhao**, Yulong Hu, Sen Li*, “RideGym: A Standardized Interface for Real-World Large-Scale Ride-Sharing System” (under review, 2027 ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)) [[ArXiv](#)] [[Code](#)] [[PyPI](#)]
- [P10] **Zijian Zhao**, Dian Jin, Xialiang Tong[#], Sen Li*, Mingxuan Yuan “Optimizing Denoising Trajectories in dLLMs: A Lightweight Evolutionary Heuristic Approach” (under review, 2027 AAAI Conference on Artificial Intelligence (AAAI)) [[ArXiv](#)] [[Code](#)]
- [P11] **Zijian Zhao**, Jing Gao*, Sen Li, “Bridging MARL to SARL: An Order-Independent Multi-Agent Transformer via Latent Consensus” (under review, 2027 AAAI Conference on Artificial Intelligence (AAAI)) [[ArXiv](#)] [[Code](#)]
- [P12] **Zijian Zhao**, Yitong Shang, Sen Li*, “AutoFed: Personalized Federated Traffic Prediction via Adaptive Prompt” (under review, 2027 AAAI Conference on Artificial Intelligence (AAAI)) [[ArXiv](#)] [[Code](#)]
- [P13] **Zijian Zhao**, Sen Li*, “Aggregate in the Advantage, Not the Ratio: A Canonical-Form Analysis of Cooperative Multi-Agent Policy Optimization” (in preparation, 2027 International Conference on Learning Representations (ICLR)) [[ArXiv](#)] [[Code](#)]
- [P14] **Zijian Zhao**, Sen Li*, “Sensing Globally, Aggregating Locally: A Scalable Reinforcement Learning Framework for Networked System” (in preparation)

D. Reports:

- [R1] Dian Jin*, Weihao Cheng, Junze Yuan, **Zijian Zhao**, “Every Author as Every Author”, 2026 [[PDF](#)] [[Code](#)]
- [R2] **Zijian Zhao**, Xuming Zhang, Jiayu Wen, Mingwen Liu*, Xiaoteng Ma, “Label Unbalance in High-Frequency Trading”, 2025 (reported by [QuantML](#)) [[ArXiv](#)] [[Code](#)]
- [R3] **Zijian Zhao***, “A Short Overview of Multi-Modal Wi-Fi Sensing”, 2025 [[ArXiv](#)] [[Code](#)]
- [R4] Dian Jin*, **Zijian Zhao**, Zijing Zhou, “IfWorld: A Multi-Agent Framework for Cross-Disciplinary Counterfactual Scenario Reasoning”, 2025 [[PDF](#)] [[Code](#)]
- [R5] Tingwei Chen, Jiayi Chen, **Zijian Zhao**, Haolong Chen, Liang Zhang*, Guangxu Zhu*, “First Token Probability Guided RAG for Telecom Question Answering”, 2025 [[ArXiv](#)]
- [R6] **Zijian Zhao**, Zitao Zhang, Kai Huang*, “A Trajectory-Based Reinforcement Learning Approach for Autonomous Locomotion of a Rat Robot”, 2024 [[PDF](#)] [[Code](#)]

Patents

A. Granted Patents:

- [1] **Zijian Zhao**, Guangxu Zhu, Qimei Chen, Kaifeng Han, "Method for Object Recognition Using Model Based on Few-Shot Learning and Related Equipment", Chinese Patent, Patent Number: ZL202411074110, 2024 [[PDF](#)]
- [2] **Zijian Zhao**, Kaifeng Han, Qimei Chen, Guangxu Zhu, Xiaoyang Li, Hang Li, "Channel State Information Recovery Method and Apparatus, Equipment, Storage Medium", Chinese Patent, Patent Number: ZL2024102321250, 2024 [[PDF](#)]

B. Published Patents:

- [1] **Zijian Zhao**, Guangxu Zhu, Kaifeng Han, Xiaoyang Li, Hang Li, "Method for Classifying Data Using Model Based on Few-Shot Learning and Related Equipment", Chinese Patent, Application Number: 2024108392137, 2024 [[PDF](#)]
- [2] **Zijian Zhao**, Guangxu Zhu, Shen Chao, Shi Qingjiang, Han Kaifeng, "Personnel Detection Method, Device, Electronic Equipment, and Storage Medium", Chinese Patent, Application Number: 2024105419689, 2024 [[PDF](#)]
- [3] Kai Huang, Zitao Zhang, **Zijian Zhao**[#], Ruoyi Tao, "A Motion Control Method for Small Bionic Rat Based on Reinforcement Learning", Chinese Patent, Application Number: 2023116499786, 2023 [[PDF](#)]

Projects

A. Projects as Student Participant (PI: Supervisor):

- [1] "Development of a Dynamic Itinerary Planning and Traffic Prediction System for Tourist Coaches", Hong Kong Society for Transportation Studies & Hong Kong Tourism Association Limited & LimX Dynamics Project, 2024-2025 [[Code](#)]
- [2] "Indoor Human Activity Recognition Based on Wi-Fi", Shenzhen Research Institute of Big Data Project, 2023-2024 [[Code](#)]

- [3] "Spectrum Efficiency Modeling with Measured MIMO Channel", Huawei & Shenzhen Research Institute of Big Data Project, 2023-2024 [\[Code\]](#)

B. Projects as Principal Investigator (Self-Initiated):

- [1] "Piano Music Generation Based on BART", College Students' Innovation Training Program at Sun Yat-Sen University, 2023-2024 [\[Paper\]](#) [\[Code\]](#)

Main Awards & Grants

A. Awards:

- [1] The Hong Kong University of Science and Technology RedBird PhD Award, 2024 (also won scholarship in 2026)
- [2] Runner Up Prize (No.2 out of 776 teams) in AI/ML for 5G-Energy Consumption Modeling by ITU AI/ML in 5G Challenge, 2023
- [3] Third Prize (No.6 out of 287 teams) in The First Wi-Fi Sensing Contest by Huawei, 2023
- [4] Best Paper Award Finalist of Biomimetics in IEEE International Conference on Robotics and Biomimetics (ROBIO) 2023, 2023
- [5] Provincial Second Prize in SPSS University Contest in Modeling, 2022
- [6] Second Prize in Asia and Pacific Mathematical Contest in Modeling, 2022
- [7] First-Class Scholarship for Outstanding Student of Sun Yat-sen University, 2021 (also won scholarship in 2022, 2023)
- [8] Meritorious Winner in the Mathematical Contest in Modeling, 2021 (also won the award in 2022)
- [9] Provincial First Prize in the Chinese Mathematics Competitions, 2021
- [10] Second Prize & Provincial First Prize in the National High School Mathematics League, 2019

B. Grants:

- [1] Student Travel Awards by ACM SIGMM, 2026
- [2] Research Travel Grant by The Hong Kong University of Science and Technology (HKUST), 2026
- [3] PhD Student Travel Fund by Hong Kong Society for Transportation Studies (HKSTS), 2026

Talks & Presentation

- [1] "Discriminatory Order Assignment and Payment-Setting of On-Demand Food-Delivery Platforms: The Impacts of Data Privacy Regulations", 2026 Huawei 2012 Labs Squirrel Seminar, 2026, Online; 2025 Bridging Transportation Researchers (BTR) Conference, 2025, Online; and 2025 KKHTCNN Symposium on Civil Engineering, 2025, Hong Kong, China
- [2] "A Lost Packet Recovery Technique for Robust Wireless Sensing", 7th IMT-2030 (6G) Plenary Meeting of Wireless Technology Group, 2024, Shanghai, China [\[PDF\]](#)

Professional Services

1. Society Membership:

- Computer Society: IEEE Student Membership, ACM Student Membership, AAAI Student Membership, CCF Student Membership

2. Conference Chair:

- Workshop Chair: ICISCC 2026 Workshop

3. TPC Membership:

- Computer & Communication Conferences: IEEE WCNC Workshop 2024-2026, IEEE PIMRC Workshop 2024-2025, IEEE GLOBECOM Workshop 2025, IEEE/CIC ICCV Workshop 2025

4. Technical Reviewer:

- Journals & Books: IEEE TPAMI, IEEE TMC, IEEE/ACM TON, IEEE TCOM, IEEE IOT-J, IEEE OJCOMS, IEEE WCL, IEEE CL, Transportation Research Part E, Frontiers in AI, MTAP, Book Manuscript from Bentham Science
- Computer & Communication Conferences: IEEE IJCNN 2025-2027, AAAI 2026-2027, IEEE ICASSP 2024-2026, IEEE ICME 2024-2026, IEEE WCNC 2024-2026, ICLR 2025-2026, ACL ARR 2025-2026, IEEE MLSP 2025-2026, NeurIPS 2026, ICML 2026, CVPR 2026, ACM ICMR 2026, ICISCC 2026, IEEE PIMRC 2024-2025, IEEE AVSS 2025, IEEE GLOBECOM 2025, IEEE/CIC ICCV 2025, IET IRC 2025, IEEE SMC 2023
- Transportation Conferences: HKSTS 2024-2025, BTR 2025

Teaching Activities

1. Teaching Assistant:

- HKUST: CIVL 4640 Introduction to Smart City Economics (Undergraduate, Spring 2026)

2. Interview:

- HKUST: JUPAS 2025

Skills & Interests

1. Programming Skills:

- Proficient in: Python, PyTorch, C/C++ (CCF-CSP:320, Top 0.8%)
- Familiar with: MySQL, Git, Linux, ESP32
- Knowledgeable in: TensorFlow, Matlab, Java, Web Scraping, QT

2. Language:

- English (IELTS:6.5, CET-4:605, CET-6: 561)
- Chinese (mother tongue)

3. Interests:

- Proficient in: Electric Guitar, Acoustic Guitar, Keyboard
- Familiar with: Electric Bass, Piano, Hulusi, Ukulele, Extreme Vocals, Music Theory
- Knowledgeable in: Drums, Harmonica

References

Prof. Sen Li: Associate Professor; Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology; E-mail: cesli@ust.hk

Prof. Guangxu Zhu: Deputy Director & Senior Research Scientist & Adjunct Associate Professor; Shenzhen Research Institute of Big Data, The Chinese University of Hong Kong (Shenzhen); E-mail: gxzhu@sribd.cn

Prof. Xiaoyang Li: Assistant Professor; Department of Electrical and Electronic Engineering, Southern University of Science and Technology; E-mail: lixy@sustech.edu.cn